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Multi-Object Ship Tracking and Predictive Path Modeling for Maritime Surveillance

A Major Project Report (CS481P)

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In partial fulfillment of the requirements for the degree of

Bachelor of Engineering in

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RV College of Engineering®, Bengaluru
(Autonomous institution affiliated to VTU, Belagavi)
Department of Computer Science and Engineering



CERTIFICATE

Certified that the major project (CS481P) work titled ***Multi-Object Ship Tracking and Predictive Path Modeling for Maritime Surveillance*** is carried out by **Shashank Shenoy B (1RV22CS183)**, **Shreyashwini R (1RV22CS192)** and **T Vinay (1RV22CS215)** who are bonafide students of RV College of Engineering, Bengaluru, in partial fulfillment of the requirements for the degree of **Bachelor of Engineering in Computer Science and Engineering** of the Visvesvaraya Technological University, Belagavi during the year 2026-27. It is certified that all corrections/suggestions indicated for the Internal Assessment have been incorporated in the major project report deposited in the departmental library. The major project report has been approved as it satisfies the academic requirements in respect of major project work prescribed by the institution for the said degree.

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DECLARATION

We, **Shashank Shenoy B**, **Shreyashwini R** and **T Vinay** students of eighth semester B.E., Department of Computer Science and Engineering, RV College of Engineering, Bengaluru, hereby declare that the major project titled '**Multi-Object Ship Tracking and Predictive Path Modeling for Maritime Surveillance**' has been carried out by us and submitted in partial fulfilment for the award of degree of **Bachelor of Engineering in Computer Science and Engineering** during the year 2026-27.

Further we declare that the content of the dissertation has not been submitted previously by anybody for the award of any degree or diploma to any other university.

We also declare that any Intellectual Property Rights generated out of this project carried out at RVCE will be the property of RV College of Engineering, Bengaluru and We will be one of the authors of the same.

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ABSTRACT

Maritime surveillance plays a crucial role in the protection of coastal security, vessel traffic management as well as the protection of marine ecosystems. The traditional systems, including radar, AIS and manual observation, are not scalable, accurate and real-time analysis and are also worse when faced with adverse conditions such as occlusion, different illumination, and even reflection on the water surface. They can not simultaneously monitor a number of vessels and predict the futures path, so they require an automated, vision-driven intelligent surveillance system.

This work introduces an end-to-end AI-based maritime surveillance pipeline with three central aims: correct multi-ship recognition, continuous identity-preserving tracking, as well as the future path prediction. YOLOv8 is a real-time, per-frame, ship detector that uses bounding box regression. DeepOCSORT is an appearance-based Re-Identification (ReID) algorithm that combines a Kalman filter state estimator with appearance-based Re-Identification (ReID). The centroid trajectories extracted are processed in an LSTM network to predict future vessel positions, with motion parameters (speed, heading, and cardinal direction) being analytically computed using the inter-frame centroid displacement.

The implementation of this system involves Python and uses PyTorch, OpenCV, NumPy, and Matplotlib to execute the system with CUDA-accelerated execution of the system on the GPUs to ensure a real-time throughput of the system. Assessment of maritime video sequences with dense traffic, occlusiveness and illumination variation ensured strong-detection, high-continuity of tracking, and consistent- multi-step LSTM-trajectory forecasting across all test cases. Integrated Circuit (IC)

Since it is a software-based system, hardware emulation cannot be used; however, the deployment-level validation on a physical GPU environment verified the real-time operational performance. The modular architecture provides a base to scale to multiple camera inputs and integrate with existing AIS and IoT infrastructure, setting a foundation to deploy maritime surveillance at production scale.

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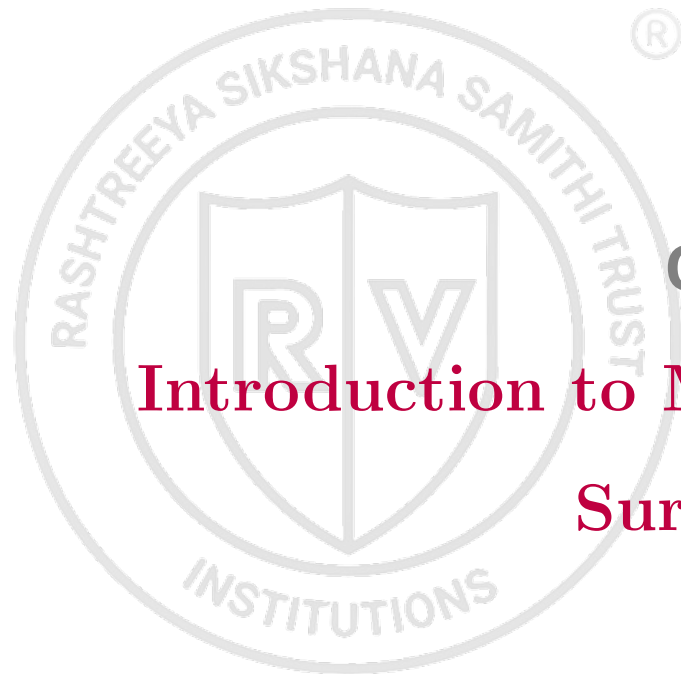
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Chapter 1

Introduction to Maritime Surveillance

CHAPTER 1

INTRODUCTION TO MARITIME

SURVEILLANCE

Maritime surveillance is the area where artificial intelligence, computer vision, and the safety of humans, countries, and the preservation of marine ecosystems are directly dependent on the technological innovation. This chapter provides the background of the project by putting into context the increasing ineffectiveness of conventional vessel monitoring solutions and explaining why a more intelligent, automated approach is needed. The motivation, problem statement, objectives, methodology and constraints of the proposed system are given sequentially, developing a logical reasoning behind the work done. Together, these sections provide the context and importance of the project, before the following chapters roll into the technical details of the project.

1.1 Introduction

The oceans and waterways of the world are the foundation of world trade, national defence, and ecological balance. More than 90 percent of international trade is transited by sea, and sea areas of the globe are home to thousands of vessels operating at the same time - commercial cargo ships and fishing boats, naval vessels, and in some areas, unauthorized or non-cooperative craft. The safety, legality and efficiency of this massive maritime activity requires sustained, precise and scalable surveillance- a challenge that has only increased significantly over the last twenty years as this vast maritime activity has blossomed. Traditionally, maritime surveillance has been based on a set of radar systems, Automatic Identification Systems (AIS), and manual video monitoring. Although these technologies have worked reasonably well in the lower-traffic operating conditions of the past, they have steadily become evidently limiting as the demands of operations have increased. Radar has detection range, but does not give any visual or contextual information on the vessels it identifies. AIS, despite its popularity, relies solely on voluntary activation of transponders and thus it is not aware of non-cooperative or unauthorized vessels. Manual video surveillance, in its turn, already is limited by human attention, fatigue, and sheer scale of maritime areas that cannot practically be covered by operators alone. Together, these strategies are reactive in nature - they react to things and not to anticipate things. With the advent of deep learning and computer vision,

the world of intelligent surveillance has entered into a totally new phase. Convolutional neural networks that can recognize objects in real time, multi-object tracking algorithms that can maintain consistent identities of objects within cluttered and dynamic scenes, and recurrent neural networks that can learn temporal movement patterns are all a good starting point of a new generation of maritime monitoring systems. When combined into a consistent end-to-end pipeline these technologies promise the potential of a system that does not just observe but also anticipates a system capable of following every visible vessel, retaining awareness of its identity and movement history, and predicting where it is heading even before it gets there. This project is called Multi-Object Ship Tracking and Predictive Path Modeling of Maritime Surveillance, and this project proposes just such a system. The system will fill the major gaps created by traditional surveillance strategies. The work is based on its pragmatic relevance - contributing to coastal security, management of port traffic, detection of illegal activities, protection of marine ecosystems - and it is consistent with the United Nations Sustainable Development Goals SDG 9 (Industry, Innovation and Infrastructure) and SDG 14 (Life Below Water) and is reflective of its greater social relevance beyond the technical domain. Figure. 4.1 below shows an image of modern AIS used in maritime surveillance.

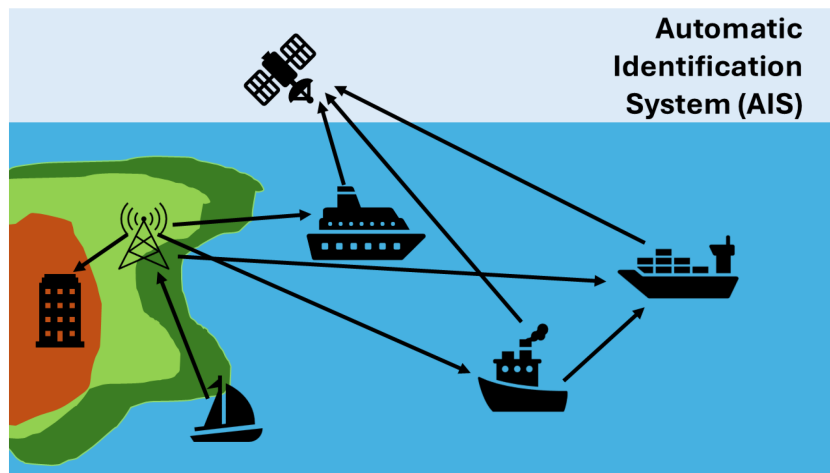


Figure 1.1: AIS System used in Maritime Surveillance

1.2 Literature Review

Maritime surveillance is one area where the last few years have seen a surge of research activity due to the increasing inadequacy of traditional methods of maritime surveillance and the rapid maturation of deep learning and computer vision technologies. The

literature reviewed covers ship detection architecture, multi-object tracking algorithm, adverse-condition robustness, and identification of gaps that drive the proposed integrated system. The next paragraphs sum up the main findings and limitations of the respective works reviewed.

Hao et al. suggested a light multi-object ship tracking algorithm, which uses an enhanced YOLOv7tiny detection backbone coupled with a StrongSORT based tracking backbone augmented by trajectory association techniques. The experiment has shown better detection accuracy with much smaller model size, which is appropriate in terms of resource constraints of the hardware. The trajectory association further minimized errors on identity switching as opposed to standard tracking baselines. The use of YOLOv7tiny instead of the more recent YOLOv8 architecture also limits the possible detection performance and the system has less robustness to situations where there is severe occlusion between closely clustered vessels.

Zhang et al. provided a detailed review of the convolutional and transformer-based solutions to visual ship detection and tracking, synthesising the development of convolutional and transformer-based image processing across different maritime scenarios. The paper has identified key issues that keep reoccurring throughout the literature and these included: occlusion, highly variable ship scales across the ranges of detection and performance degradation in harsh weather conditions, such as, fog, rain and low light conditions. Although the survey does give a good understanding of the state of the art, it does not suggest an original integrated solution or real-time implementation, which limits its direct applicability as a deployable solution.

Li and Wang created EGM-YOLOv8, a lightweight version of the YOLOv8 detection model with an efficient feature fusion module and channel attention mechanism specifically designed to detect maritime ships. The model also attained a 13.57% reduction of parameters relative to the standard YOLOv8 baseline and at the same time, it is also able to achieve a greater recall across test scenarios hence demonstrating excellent real-time performance suitable on resource limited surveillance platforms. Although these contributions are made, the work is about the detection in isolation - no tracking and trajectory prediction module is included, and the utility of the system is limited to the single-frame localization, but without the tracking and trajectory prediction module, and without the behavioural analytics.

Jian et al. developed a maritime ship target detection study, through an optimization of the YOLOv8 model, which has been trained on a large scale dataset of over 80 thousand annotated maritime images, with a diverse range of sea conditions, vessel orientation and environmental factors. The paper found that performance on complex sea scenes was significantly enhanced by large-scale domain-specific training data, over baselines not optimised. Nevertheless, the work is limited to detection, but no one thinks about the multi-frame tracking, identity preservation, or movement prediction. The optimization cost of the optimized model is also reported to be more than the lightweight alternatives, a factor that is also of concern when it comes to the implementation of the optimized model in real-time embedded systems.

Ma et al. suggested DeepSORT-OCR a ship tracking system based on DeepSORT, which adds an Optical Character Recognition (OCR) module that could read hull identification numbers on the imagery of the vessels. The system was able to recognize Multi-Object Tracking Accuracy (MOTA) of 77.18% and show a much higher identity consistency compared to appearance-only tracking baselines. In spite of these advantages, the scheme has a very serious operational disadvantage, namely, it must have visible hull numbers, and it must have high-resolution camera images, which makes the scheme unworkable in long-range surveillance scenarios, or in situations where hull marks are obscured by weather, distance, or the angle of the camera.

Yuan et al. created RT-FogNet, a real-time ship detection model that is specifically designed to operate under low visibility conditions in the maritime system, such as fog, backlight, and water surface reflections in the inland waterways. The model utilises a lightweight platform that is optimised to run on realistic waterway infrastructure, and can deliver reliable detection performance under less than ideal illumination conditions under which conventional detectors fail. Although RT-FogNet is a significant step towards robustness to environmental degradation, the system is designed to work only in the inland waterway settings with only one object to be tracked, and does not address the open-sea multi-object tracking, vessel identity management or predictive path modeling.

1.2.1 Review of Literature

The literature surveyed for this project spans four key domains: ship detection using deep learning architectures, multi-object tracking algorithms for maritime environments, trajectory prediction using time-series models, and robustness under adverse maritime

conditions. The reviewed works collectively highlight the evolution of AI-driven surveillance techniques and reveal critical gaps in existing systems, particularly the absence of integrated end-to-end pipelines combining detection, tracking, and predictive modeling.

Ship Detection in Maritime Environments

Recent research has demonstrated significant progress in applying convolutional neural network-based detectors to maritime ship detection. Li and Wang Li2025 proposed EGM-YOLOv8, a lightweight variant of YOLOv8 enhanced with efficient feature fusion and channel attention mechanisms, achieving a 13.57% reduction in model parameters while improving recall for maritime ship detection. Jian et al. Jian2025 developed an optimized YOLOv8 model trained on a large-scale dataset of over 80,000 annotated maritime images, demonstrating improved detection accuracy across complex sea scenes with varying vessel orientations and environmental conditions. Yuan et al. Yuan2026 introduced RT-FogNet, a real-time detection model specifically engineered for low-visibility inland waterway conditions including fog, backlight, and water surface reflections, achieving reliable detection under adverse illumination where conventional detectors fail.

Multi-Object Tracking for Vessel Monitoring

The consistent tracking of multiple vessels across dynamic video frames presents unique challenges in maritime environments, including occlusion, re-entry, and cluttered backgrounds. Hao et al. Hao2025 proposed a lightweight multi-object ship tracking algorithm combining an improved YOLOv7tiny backbone with StrongSORT-based tracking and trajectory association, reducing identity switching errors and demonstrating suitability for real-time maritime use on resource-constrained hardware. Ma et al. Ma2026 presented DeepSORT-OCR, which augments the standard DeepSORT tracker with an OCR module for hull number recognition, achieving a Multi-Object Tracking Accuracy (MOTA) of 77.18% with stronger identity consistency compared to appearance-only tracking baselines. Zhang et al. Zhang2025 provided a comprehensive survey of deep learning methods for visual ship detection and tracking, cataloguing recurring challenges such as variable ship scales, occlusion, and performance degradation under harsh weather, forming a strong conceptual basis for the design of the proposed system.

Research Gaps Identified

A critical synthesis of the reviewed literature reveals several persistent gaps that motivate the proposed work. The majority of existing systems address only one component of the surveillance pipeline — either detection or basic tracking — without integrating both into a cohesive real-time framework. None of the reviewed works combine ship detection, identity-preserving multi-object tracking, motion parameter estimation, and future trajectory prediction within a single end-to-end system. Tracking systems dependent on hull number recognition introduce operational constraints unsustainable under real-world maritime conditions, while detection-only models provide no temporal awareness of vessel behaviour. Furthermore, predictive path modeling from video-based inputs remains largely unaddressed in the current literature. The proposed system directly addresses these gaps by integrating YOLOv8-based detection, DeepOCSORT tracking with ReID, LSTM trajectory prediction, and geospatial visualization into a unified real-time maritime surveillance pipeline Hao2025, Zhang2025, Li2025, Jian2025, Ma2026, Yuan2026.

1.3 Motivation

The surveillance of the sea is one of the most severely under-serviced aspects of contemporary security and environmental surveillance even though its practical importance is immense. There are thousands of vessels in operation at the same time in vast oceanic areas, but the instruments at their disposal to keep track of them are, for the most part, archaic and unsatisfactory. Radar systems do not provide any visual context, AIS can only be scaled to the breadth of the geographic spread of actual maritime activity. What was of special interest to us in this problem was its multi-layered character—it was not sufficient to detect ships, but the system had to continually track each ship across frames, manage occlusions and reappearances, and predict future motion, all in real time, under random and unpredictable lighting and sea conditions. On top of a technical challenge, the cost of poor maritime surveillance is very tangible in terms of accidents, illegal fishing, smuggling, and border violations all come with real costs to human safety, national security and marine biodiversity. This project was both technically interesting and truly valuable, which is why it aligns with SDG 9 (Industry, Innovation and Infrastructure) and SDG 14 (Life Below Water).

1.4 Problem statement

The current maritime surveillance systems, radar, AIS and manual video surveillance are all found wanting in their ability to provide vessel oversight over large maritime zones and to provide a visual context, to be dependent on vessel compliance and to scale to large maritime areas. None of these methods can identify multiple vessels simultaneously, have identities consistent across dynamic video frames or predict future trajectories and authorities can always be reactive. The environmental issues like different sources of illumination, reflections by water, and occlusiveness also worsen their reliability. Consequently, there is an urgent requirement of an automated, vision-based intelligent system that can be able to detect multiple ships in real-time, conduct persistent identity-preserving tracking, and predictive path modeling to allow it to perform proactive maritime surveillance and make informed decisions across the coastal security, port management, and marine ecosystem protection.

1.5 Objectives

The objectives of the project are

1. To develop an intelligent maritime surveillance system capable of detecting and tracking multiple ships in real-time using YOLOv8 for accurate ship detection and DeepOCSORT for maintaining consistent vessel identities across frames, robust against real-world maritime challenges such as occlusion, lighting variations, and water surface reflections.
2. To extract trajectory information from tracked vessels and apply predictive models — specifically Long Short-Term Memory (LSTM) networks and Kalman Filters — to forecast future ship paths, enabling proactive monitoring and early detection of anomalous vessel behaviour.
3. To design an integrated visualization system that displays detection results, unique tracking identifiers, movement trajectories, predicted paths, and geospatial vessel positions, providing a comprehensive real-time situational awareness interface for maritime surveillance applications.

1.6 Brief Methodology of the project

The system proposed has a modular pipelined architecture in which each stage processes and sends data to the next, making it scalable and able to perform in real time. Maritime video ingestion initiates the methodology that involves the extraction and pre-processing of raw frames into model input. Every frame is processed through a YOLOv8 detection model to detect and localize all ships using a bounding box regression model. Detected ships are then inputted into the DeepOCSORT multi-object tracking module which assigns and maintains unique identities across frames by appearance-based Re-Identification (ReID) and Kalman filter state estimation. Over time centroid coordinates of tracked vessels are recorded and trajectory sequences of the tracked vessel built by centroid coordinates and motion parameters, such as speed, heading, and cardinal direction, are derived analytically using inter-frame displacement. These paths-sequences are then used by an LSTM neural network to predict the future vessel positions. Lastly, all outputs (bounding boxes, tracking IDs, motion vectors and predicted paths), are displayed on the video feed and geospatially mapped with Folium, to provide a single real-time interface of maritime monitoring.

1.7 Assumptions made / Constraints of the project

The system is planned, and tested on the following assumptions. Input is presumed to be video footage captured by a fixed camera in one location; a coastal watchtower or a port surveillance unit. It is assumed that ships are sufficiently visible and distinguishable to the water background and that the YOLOv8 model has been trained on a representative maritime dataset that covers the variations in ship size, type and environmental conditions. The movement of vessels is also assumed to be governed by time-invariant trajectories since sharp manoeuvres can decrease the prediction accuracy of LSTMs. It is also assumed that a computing environment with a graphics card is available since real-time throughput cannot be assured on hardware without graphics cards. As follows, the constraints control the extent and legitimacy of the project. The system only accepts video as an input and does not take radar or AIS data. The performance of detection and tracking can diminish when the weather is extreme, i.e. when the visibility is low and the rain is heavy. The accuracy of prediction of LSTM is limited by the length and quality of data available on the historical trajectory of a given vessel, as well as the fact that the accuracy of prediction decreases with longer and lower quality data available

on the historical trajectory of a given vessel. Single-feed surveillance is the only type that is currently supported by the system, and it does not accept several simultaneous camera feeds. The accuracy of geospatial mapping process also requires accurate camera calibration parameters to ensure accurate pixel to coordinate mapping.

1.8 Organization of the Report

This report is organized as follows.

- Chapter 2 discusses the review of existing literature on maritime surveillance systems, covering ship detection, multi-object tracking, and trajectory prediction methods, along with identified research gaps that motivate the proposed system.
- Chapter 3 discusses the system design and methodology, elaborating the modular pipeline architecture encompassing video ingestion, YOLOv8-based detection, DeepOCSORT tracking, motion parameter estimation, LSTM trajectory prediction, and geospatial visualization.
- Chapter 4 discusses the implementation details of each module, including dataset preparation, model training procedures, integration of the prediction pipeline, and the tools and frameworks employed for development.
- Chapter 5 discusses the experimental results and performance evaluation of the system, presenting detection accuracy, tracking consistency, trajectory prediction outcomes, and qualitative visualization outputs across various maritime test scenarios.
- Chapter 6 discusses the conclusions drawn from the project, highlighting key contributions, limitations of the current system, and potential directions for future enhancement and real-world deployment.

Chapter 2

**Review of existing literature on
maritime surveillance systems,
covering ship detection, multi-object
tracking, and trajectory prediction
methods**

CHAPTER 2

REVIEW OF EXISTING LITERATURE ON MARITIME SURVEILLANCE SYSTEMS, COVERING SHIP DETECTION, MULTI-OBJECT TRACKING, AND TRAJECTORY PREDICTION METHODS

This fast development of deep learning and computer vision has changed the way maritime surveillance research is conducted, and a series of studies on intelligent, automated solutions to traditional radar and manual surveillance systems have emerged. The chapter reviews the literature on the three main technical aspects of the proposed system, namely, ship detection, multi-object tracking, and trajectory prediction, from the latest publications in the most prestigious journals such as Expert Systems with Applications, Engineering Applications of Artificial Intelligence, Journal of Marine Science and Engineering and Scientific Reports. The characteristics of each reviewed work are analyzed for their major contributions and methodological decisions as well as their limitations, and the works are specifically analyzed in terms of how well they address the complexities of the real world in maritime surveillance. The gaps as a whole that have been identified from literature are the technical and operational justification for the end-to-end integrated system proposed in this project.

2.1 Details about the Literature Review of the papers conducted for this project

The current studies on maritime surveillance based on deep learning show a fragmented and evolving literature, with each study tackling a particular aspect of the overall surveillance tasks without providing a coherent end-to-end approach. Although the proposed multi-object ship tracking system is lightweight and competitive to other tracking systems in real-time, it is still limited to ship detection performance due to using an improved YOLOv7tiny backbone, and it shows poor robustness in extreme occlusion while has no predictability at all. Although highly useful as a reference, Zhang et al. suggest no novel system, and do not even touch on the integration of detection,

tracking, and prediction, which remains the main hurdle in the application of deep learning to real-world maritime surveillance, and the key challenges of the maritime domain: occlusion, extreme scale variation, water surface clutter, and adverse weather degradation. Although the model parameters were reduced by 13.57% compared to YOLOv8, EGM-YOLOv8 was still lightweight, showing excellent real-time detection performance in maritime scenes, especially on small and distant targets, it only performed detection tasks and lacked temporal awareness, identity tracking, and trajectory analysis in the subsequent video frames. While Jian et al. have contributed to the performance gains in complex sea scenes with high detection accuracy, and generalisation across a variety of maritime scenes and states, the system's restrictions on computation and single-frame detection (with no tracking, trajectory extraction, or predictive module) severely limit its applicability to the most basic stage of the surveillance pipeline. To address the issue, Ma et al. developed a framework called DeepSORT which is extended by adding an OCR module to extract hull identification number as an additional hull feature to be used for re-identification, but a high resolution image is needed for the system to work, the hull number needs to be visible, and it cannot be used for long-range surveillance or in adverse weather conditions, and there is no trajectory prediction function. The system is designed for low visibility inland waterway conditions (such as fog, backlight and water reflections) and is based on dehazing-aware feature extraction and illumination-robust normalisation, making it robust even in foggy scenarios but limited to inland waterway detection scenarios, without any tracking, identity management, motion analysis or path prediction. A common and critical deficiency identified in all reviewed systems is the lack of a real-time, unified pipeline that can perform ship detection, persistent multi-object tracking with identity-preserving, motion parameter estimation and future trajectory prediction — a deficiency which fuels and sets the boundaries of the system proposed in this project.

2.2 Programming Language used

The suggested maritime surveillance system is developed in full using Python, a high-level interpreted language which now is the de facto standard for Artificial Intelligence, Deep Learning and Computer Vision research and development. The simplicity and eloquence of Python's syntax are also driving its use in these applications — by streamlining the process of coding mathematically intensive algorithms, engineers can concentrate on

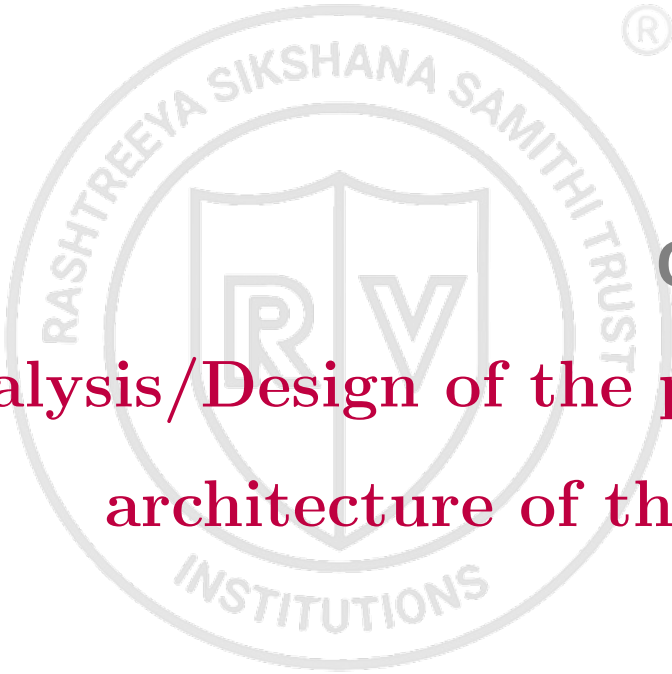
designing their systems and can model the logic of the systems without having to worry about all the implementation details. However, Python is also the native interface language of almost all current, widely-used deep learning and computer vision libraries, providing seamless interoperability between all components of the proposed pipeline. It is also the most practical and productive option for an academic project this size and interdisciplinary in scope because of its extensive community support, rich documentation and its ability to be rapidly prototyped.

2.3 Tools and Technologies used

The system is based on a properly selected set of open-source libraries and frameworks that play specific roles in the pipeline. PyTorch is the backbone of deep learning, supporting dynamic building of computational graphs, automatic differentiation and native support of the CUDA libraries for the training and inference of models on GPUs. Building on PyTorch, the Ultralytics YOLOv8 library offers an easy to use API to create ship detection models, set up training and deployment parameters, and use the model for real-time inferences. Video ingestion is done with OpenCV, a computer vision library that has been highly optimised for real-time frame-by-frame processing pipelines, needed for live surveillance applications, such as video ingestion. OpenCV is a highly optimised computer vision library, which supports frame-by-frame processing pipelines, and therefore is indispensable for live surveillance applications, including video ingestion. The DeepOCSORT tracking module is a fully integrated tracking module that fuses directly with the detection output to preserve vessel identities from frame to frame via appearance-based Re-Identification and Kalman filter state estimation. Using NumPy and Pandas, numerical computation, array manipulation, and calculations of motion parameters (centroids) are carried out, which is fast and vectorised data handling of large trajectory datasets. Trajectory and detection visualisations are created in the video output using Matplotlib, geospatial vessel position mapping is created using Folium and interactive maps of tracked vessel positions are generated. All of the pipeline is speeded up via NVIDIA CUDA, allowing the system to process the detection, tracking, and prediction tasks on real time. Testing and experimentations are done in Jupyter Notebook and Visual Studio Code, enabling iterative testing and integration of modules before rollout in the complete pipeline.

2.4 Summary

This chapter had conducted a comprehensive survey of the existing literature in maritime surveillance, systematically reviewing the published literature across the marine detection field, multi-object tracking and trajectory prediction fields from top-notch academic journals. In summary, the summarized works in this review show that deep-learning and computer vision approaches have rendered great achievements in the development of robust, lightweight detection systems, as well as appearance-based tracking models and techniques for maritime monitoring purposes which are resilience and adaptable to different adverse environmental conditions. But there is a hardly autarkic mistake which prevails throughout all above-mentioned publications: up to now there is no satisfactory system combining ship detection with keeping ships identity within a track record of call-signs or unique identifiers, motion analysis into a track, and forecasting the trajectory of the sought ships in a single real-time pipeline. The programming tools and language used in the development of this project were identified and justified as the most suitable and effective tools available to meet this gap, these tools were: Python, PyTorch, YOLOv8, DeepOCSORT, LSTM, OpenCV and Folium. The identified research gaps and the technological alternatives proposed in this chapter in respect to the maritime surveillance system, directly feed into the discussion of the overall system design and method proposed for their implementation in the subsequent chapter, which more broadly outlines the required architectural choices and design rationale behind this end-to-end maritime surveillance system.

The logo is a circular emblem. The outer ring contains the text 'RASHTREEYA SIKSHANA SAMITHI TRUST' at the top and 'INSTITUTIONS' at the bottom. In the center is a shield divided vertically, with a large 'R' on the left and a large 'V' on the right.

Chapter 3

Analysis/Design of the pipelined architecture of the system

CHAPTER 3

ANALYSIS/DESIGN OF THE PIPELINED ARCHITECTURE OF THE SYSTEM

The architecture of the proposed maritime surveillance system is designed in the form of a multi-stage, modular pipeline, where each stage serves a specific purpose, receiving enhanced data from the previous one and sending it to the next one, thereby providing a process clarity, as well as a real-time scalability. The system feeds two sets of data into the system, a continuous video stream from the drone, CCTV or satellite camera and an AIS data stream which includes vessel identifiers, positional coordinates, speed and course data. These are processed separately, then integrated in a sensor fusion layer that combines the spatial information of video-based tracking with the navigational information of the AIS telemetry. The video pathway sequentially detects ships using YOLOv8, tracks identities without loss using DeepOCSORT, and processes the data through homographic pixel-to-coordinate transformation to build the geo-referencing path, whereas the AIS pathway applies frame-rate interpolation to then match with the video path using Haversine and Hungarian assignment algorithms. Enriched tracks, which include MMSI, speed, course, and ship type, are then fused with noisy video detections using a Kalman filter that reconciles the higher-frequency, but less precise, video stream with the more-sparse but more precise AIS observations to obtain a combined state estimate for the geographic position and velocity of each vessel. The resulting feature sequences (latitude, longitude, speed over ground, and trigonometric course encodings) are passed through an LSTM path predictor which predicts the future position of the vessel over the next K timesteps, followed by a post-processing stage of applying a land mask filter to ensure predicted paths remain in physically valid water areas.

3.1 Details about the system architecture

3.1.1 Specifications for the design

The maritime surveillance system being proposed is expected to meet the following technical requirements. The system can receive continuous video input with a minimum resolution of 1280×720 , and processes the video at a goal frame rate of 30 frames per second, which ensures real-time video surveillance throughput. The YOLOv8 detection

module is set to be a trade-off between detection sensitivity and suppressing false positives, with a confidence threshold of 0.5 and an Intersection over Union (IoU) threshold of 0.45 for Non-Maximum Suppression. The DeepOCSORT tracker is instantiated with the default values of the maximum age of 30 frames for the retention of the lost track, minimum hit count of 3 for the confirmation of a track, and cosine similarity as the distance measure used for appearance-based Re-Identification. The prediction module that is used by LSTM takes 30 timesteps as input and predicts the next K timesteps for the vessel's location. In the geospatial mapping module, the pixel coordinates are transformed into geographic latitude-longitude pairs with the aid of homographic transformation matrices generated with the camera calibration parameters. For real-time performance, NVIDIA CUDA acceleration is required and a minimum of 6GB VRAM is required. All processing and video ingestion operations are performed at the frame-level and the system is built using PyTorch as the main deep learning framework and OpenCV to process the frames.

Seeing as the main objective of any design is to gauge how the software will function, it is essential to perform some pre-analysis work beforehand. Considering the primary goal of any design is to analyze how the software will operate, it is necessary to undertake some pre-analysis work prior to the design.

A comprehensive pre-analysis was performed prior to the design of the system that involved reviewing the data sets, selecting a suitable model and algorithmic benchmarking so each selected component was well matched to the maritime environment.

3.1.2 Dataset Analysis

The quality of the annotations, diversity of the environment, and the proportion of vessel classes in the maritime video data were assessed. A variety of sea states, illumination conditions, vessel scales, and occlusion scenarios were selected as a priority to be included in the datasets, so as to allow the trained detection model to adapt well to the various scenarios of practical surveillance. YOLO format frame extraction and annotation were done, and bounding boxes were annotated for each instance of a vessel for training, validation and test sets.

3.1.3 Detection Model Selection

After comparing various YOLO models, YOLOv8 is chosen as the detection backbone. Compared to YOLOv5, YOLOv7 and YOLOv7tiny, YOLOv8 is selected as the

detection backbone. The anchor-free architecture, optimized feature pyramid network, and enhanced performance in detecting small objects made YOLOv8 the best option for maritime ship detection, particularly in terms of small object detection capabilities, which is crucial for identifying ships in the distance.

3.1.4 Tracking Algorithm Selection

After experiments, DeepOCSORT is chosen to be the best algorithm to preserve identities in dense, occluded maritime scenes among the two algorithms. Considering the performance in preserving identities, DeepOCSORT is compared and selected as the best algorithm for the dense maritime scenes with occlusion. When comparing the resistance to identity switching using DeepOCSORT with OSNet-based Re-Identification to other motion-based trackers, such as motion-only trackers, DeepOCSORT is shown to be the best tracker for persistent monitoring of vessels.

3.1.5 Prediction Model Selection

Instead, the trajectory prediction was done using LSTM networks, rather than Kalman Filters or GRU variants. Kalman Filters are a computationally efficient linear motion estimation but are unable to represent the non-linear manoeuvres found in the movement of maritime vessels that have curved movements. Unlike LSTMs, which try to learn complex temporal dependencies from the historical position sequences, the current approach cannot learn such dependencies. A kinematic linear regression fallback is implemented for data with less than 30 steps in the trajectory history to be used as input for the LSTM.

3.1.6 Sensor Fusion Analysis

Identified a key design opportunity for the complementary nature of video tracking with AIS telemetry. Video is good at observing high frequency, but it is noisy and subject to degradation due to environmental conditions, while AIS is inaccurate in its observations, but is sparse. Kalman filter fusion strategy was used to take advantage of both sources and combine their respective individual strengths to get a single coherent and stable estimate of the state of the vessel.

3.2 Summary

A detailed description of the design and methodology used in the proposed maritime surveillance system were presented in this chapter. From detection confidence thresholds to tracker initialisation parameters, to LSTM input window lengths to GPU requirements, the specifics of each module were set to define the boundaries of operation of the system. The pre-analysis results were presented, which helped to choose the YOLOv8, DeepOCSORT, LSTM, and Kalman filter fusion algorithms as the main components. The main components of the algorithms, such as YOLOv8, DeepOCSORT, LSTM, and Kalman filter fusion, were selected after a detailed pre-analysis, where each design decision was substantiated with comparative analysis with other algorithms. Then the two-phase pipeline architecture was developed to detail the flow of data from ingestion of raw video and AIS stream to detection, identity-preserving tracking, geo-referencing, sensor fusion, feature sequence construction to trajectory prediction, output generation with the constraint of the land-mask. The mathematical formulations were presented to give a rigorous theoretical background of the behavior of the mathematical system involved in each stage of the system, such as bounding box regression, IoU computation, Cosine similarity matching of ReID, Haversine distance calculation, Kalman filter state estimation and LSTM gating equations. Lastly, the experimental methods used to prepare the datasets, train the models and test the pipeline with the integrated system were presented, outlining the process of how the system is tested. The rest of this chapter and the subsequent chapter implement the design, which is fully described in this chapter.

Chapter 4

**Implementation of Pipelined
Maritime Surveillance and Teacking
System**

CHAPTER 4

IMPLEMENTATION OF PIPELINED MARITIME SURVEILLANCE AND TEACKING SYSTEM

The process of turning a carefully drafted system design into a working real-time surveillance pipeline requires careful attention to the integration of the different tools, frameworks, and data flows for each processing stage. The proposed maritime surveillance system integrates the entirety of the design choices made in the previous chapter, and extends them to become a fully functional system that can be tested. The proposed maritime surveillance system combines the entire design decisions that were made in the previous chapter, extends this and makes them operational and viewable in a system that can be tested. Each module was designed, configured and tested separately before being integrated into the complete end-to-end system to make sure that any errors could be detected and corrected at the module level before the integration of the entire system was made. The chapter provides a detailed description of the system implementation in practice, which includes the development environment, dataset preparation procedures, model training configurations, implementation specifics for each module, and the strategies used to obtain a stable real-time throughput, and hence, a complete and reproducible account of the realisation of the proposed design. Below is the pipeline architecture of the proposed system.

4.1 Top-Level System Design and Implementation

The proposed maritime surveillance system is a fully software-based implementation built upon a modular pipeline architecture. The top-level design integrates six primary functional modules — video ingestion, ship detection, multi-object tracking, sensor fusion, trajectory prediction, and geospatial visualisation — each implemented as an independent unit that communicates with adjacent stages through well-defined data interfaces. The following sections elaborate the implementation details of each module.

4.1.1 Development Environment and Configuration

The system is implemented in Python 3.8, with PyTorch serving as the primary deep learning framework for both the YOLOv8 detection model and the LSTM trajectory predictor. OpenCV handles all video ingestion, frame extraction, and real-time annotation

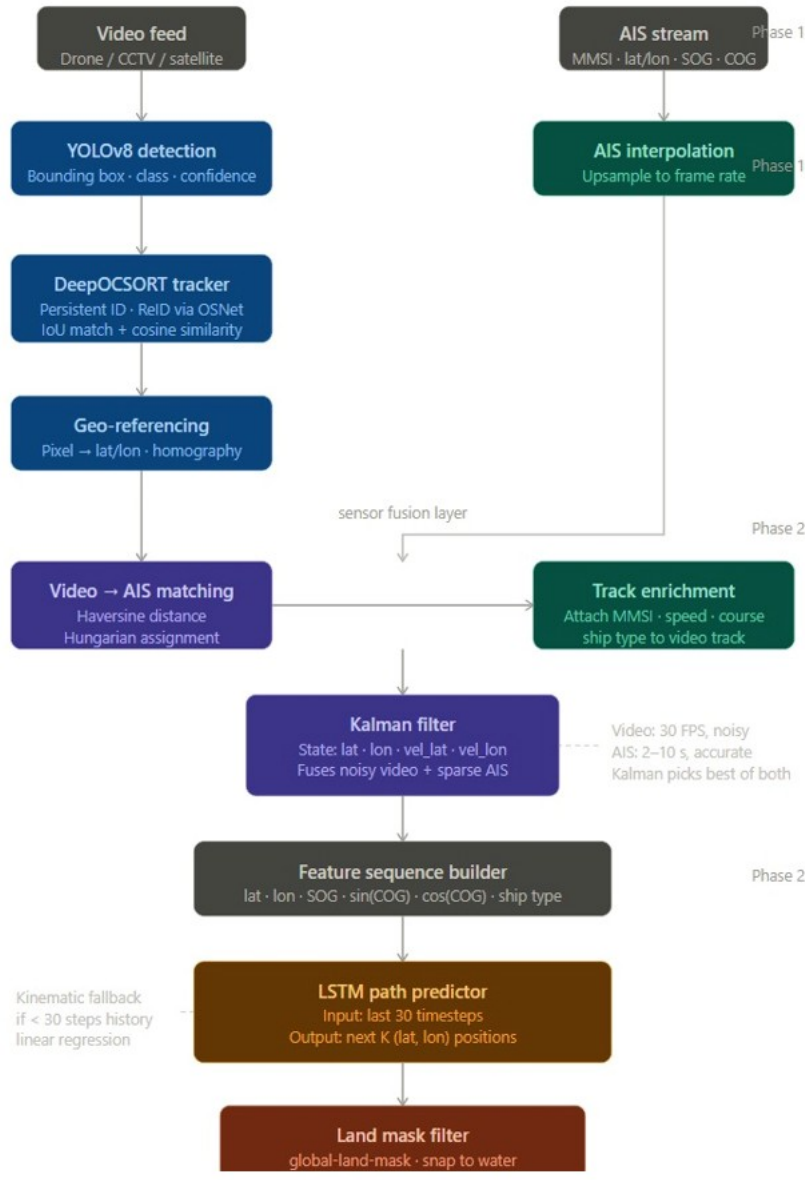


Figure 4.1: Pipeline of the proposed system

rendering. NumPy and Pandas manage intermediate data structures including bounding box arrays, centroid coordinate histories, and trajectory sequence buffers. Matplotlib provides offline visualisation of tracking outputs and predicted paths, while Folium renders interactive geospatial maps displaying vessel positions in real-world latitude-longitude coordinates. The entire pipeline is executed on an NVIDIA GPU-enabled environment with CUDA acceleration, ensuring that frame-level processing meets the 30 FPS real-time throughput requirement. Development and testing were conducted using Jupyter Notebook and VS Code as the primary integrated development environments.

4.1.2 Dataset Preparation and Annotation

Maritime video datasets were sourced and preprocessed to generate the training corpus for the YOLOv8 detection model. Raw video footage was segmented into individual frames at uniform sampling intervals, producing a diverse image dataset encompassing varied sea states, vessel types, illumination conditions, and occlusion scenarios. Each frame was annotated using bounding box labels in YOLO format, with each annotation specifying the class identifier and normalised centre coordinates, width, and height of every vessel instance present. The annotated dataset was partitioned into training, validation, and test splits following a 70:20:10 ratio. Data augmentation techniques — including horizontal flipping, brightness and contrast jittering, random cropping, and mosaic augmentation — were applied during training to improve model generalisation and robustness across unseen maritime conditions.

The Top-Level System Design and Implementation section outlines the overall design and implementation of the system at a high level. The Top-Level System Design and Implementation section describes the overall design and implementation of the system at a high level.

The proposed maritime surveillance system is a fully software-based system, based on a modular pipeline architecture. The high-level design consists of six main functional modules – video ingestion, ship detection, multi-object tracking, sensor fusion, trajectory prediction, and geospatial visualisation – with each module being a standalone entity that provides clearly defined data interfaces to connect with neighbouring modules. The next sections discuss the details of the implementation of each module.

Subsection Development Environment and Configuration The entire system is developed in python 3.8, and PyTorch is used as the main deep learning framework for the detection model (YOLOv8) and trajectory predictor (LSTM). OpenCV provides functionality to ingest all video footage, extract frames and render all the annotations in real time. Intermediate data structures such as bounding box arrays, centroid coordinate histories, and trajectory sequence buffers managed by NumPy and Pandas. Matplotlib allows for plotting tracking outputs and predicted tracks offline, and Folium generates interactive geospatial mapping with customizable features for plotting vessel positions in world coordinates (latitude and longitude). The whole pipeline is implemented in an NVIDIA GPU accelerated environment, CUDA, that guarantees the real-time through-

put of 30 FPS at the frame level. The primary integrated development environment used for development and testing were Jupyter Notebook and VS Code.

4.1.3 Data Collection and Data Annotation

All the maritime video data were collected and pre-processed to create the training data set for the YOLOv8 detection model. The raw video data was then partitioned into individual frames with a fixed sampling interval, resulting in a wide range of images representing varying sea-states, vessel types, illumination conditions and occlusion scenarios. The images were annotated using bounding box labels in YOLO format that included the class identifier and normalised centre coordinates, width and height of all the vessels instances within each image. The data set was divided into training, validation, and test sets, with ratios set as 70:20:10. To enhance the generalisation and robustness of the model for unseen maritime conditions, data augmentation techniques were used during training, such as horizontal flipping, brightness and contrast jittering, random cropping and mosaic augmentation.

The Ship Detection Module is YOLOv8. YOLOv8 is Ship Detection Module. Pre-trained COCO weights were used to initialise the YOLOv8 model and the model was fine-tuned with the prepared maritime dataset using transfer learning. The training was done with the Stochastic Gradient Descent (SGD) optimiser, the Cosine Annealing learning rate scheduler and early stopping based on the convergence of the validation mAP. The model has been configured to have a confidence threshold of 0.5 and an IoU threshold of 0.45 for Non-Maximum Suppression during inference. During runtime, for every video frame extracted, it is input into the trained YOLOv8 model that outputs a list of bounding box predictions, confidence scores, and class labels for all the detected vessels. Any detections that are not above the confidence level will be ignored and not sent to the tracking module.

The Multi-Object Tracking Module — DeepOCSORT module is used to track the objects in the image. The Multi-Object Tracking Module — DeepOCSORT module is applied to track objects in the image. Confirmed detections by the YOLOv8 module are fed to the DeepOCSORT tracker, which is initialised with a maximum track age of 30 frames, a minimum hit count of 3 for the tracker to be considered as a confirmed track, and cosine similarity metric for Re-Identification (RI). OSNet is used as the appearance embedding extractor and results in compact feature vectors of detected vessels, which are

encoding appearance identity information without position information. DeepOCSORT first matches detections to track predictions at every frame according to IoU, and then further matches the detections to the track predictions by cosine similarity using ReID embeddings in order to disambiguate the matching. Successful matches update the corresponding tracks, unmatched detections start new tentative tracks and tracks that do not get a match within the maximum age are removed. The coordinates of all tracks that have been determined as active at each frame are extracted and stored in a moving track buffer for use in downstream motion analysis and prediction.

4.1.4 Motion Parameter Estimation

The tracking module stores a history of the centroid coordinates, which is used to compute inter-frame motion parameters analytically. Instantaneous vessel speed is computed as the Euclidean distance travelled per unit time between the centroid of neighbouring frames, for each active track. The heading angle is found by taking the arctangent of the vertical and horizontal displacement components and the cardinal direction (North, South, East, West and their intermediate directions) is produced by dividing the calculated heading angle into direction bins. These motion parameters are added to the video output stream together with the Bounding Boxes and the Tracking ID's, so that the operator is given a live stream of the motion parameters for each tracked vessel.

Finally, the sensor fusion – AIS integration and Kalman filter are discussed. AIS data (with MMSI identifiers, latitude, longitude, Speed Over Ground (SOG) and Course Over Ground (COG)) is received concurrently with the video data and upsampled using linear interpolation to the video frame rate. The positions of video track are geo-referenced by applying a pre-computed homographic transformation matrix to the video centroid pixel positions, and the matched AIS records are found through Haversine distance computation and Hungarian assignment algorithm. tracks that are successfully matched will be enhanced with AIS vessel attributes. A Kalman filter is then applied to the fused state, with the state vector containing latitude, longitude and velocity in the latitude and longitude directions. The Kalman filter prediction and update steps combine the high-frequency video observations, which are noisy and uncertain, with the limited (but high accuracy) AIS observations, to get a smooth and stable estimate of each vessel's geographic position at each timestep.

4.1.5 Trajectory Prediction Module — LSTM

The LSTM is used to predict the trajectory in the Trajectory Prediction Module. In the Trajectory Prediction Module the LSTM is used to predict the trajectory. The LSTM trajectory predictor is a sequential neural network, consisting of two stacked LSTM layers and a fully connected output layer. The model is trained on input sequences of data of length 30 timesteps, each timestep being a feature vector composed of latitude, longitude, SOG, sin(COG), cos(COG) and ship type encoding data. The network was trained on historical track data using a loss function known as Mean Squared Error (MSE) and the Adam optimiser. During the runtime the feature sequence builder constructs the 30-step window for the input data, from the history of the track of the Kalman-filtered vessels, and passes it to the trained LSTM model, which returns the predicted geographic coordinates for the next K positions of the vessels. To achieve this, the kinematic linear regression fallback is automatically enabled for tracks with less than 30 timesteps of history, allowing for track output whether or not the track is new.

4.1.6 Land Mask Filter and Geospatial Visualisation

The predicted trajectory coordinates are sent to a Land mask filter which compares each predicted point with a global land-water mask to check if it is either in land or water. Predictions that fall on land areas that are indicative of physically unrealistic vessel routes due to prediction uncertainty are snapped to the nearest valid water coordinate to limit all predictions to operationally realistic maritime areas. Detection bounding boxes, unique tracking ID, motion-parameter annotations, historical trajectory overlay and predicted path markers are drawn on the live video stream by OpenCV. At the same time, the positions of vessels and potential paths are displayed on an interactive Folium map, delivering to the boat operator a geospatial, real-time dashboard for all maritime situational awareness. You can add sections and sub sections to elaborate your project work done.

4.2 Summary

This chapter fully implemented the proposed end-to-end maritime surveillance system, and this system was based on the architectural design developed in Chapter 3, to enable the technology to be fully operational and real-time. The development environment was

defined and the technology stack consists of Python, PyTorch, OpenCV, NumPy, Pandas, Matplotlib and Folium, with the processing power and acceleration of the system provided by NVIDIA CUDA. The data preparation and data annotation process were explained, including extraction of frames from the video and labelling the YOLO format, partitioning of the data and methods used to augment the data to increase model generalisation. Then the implementation of each functional module was elaborated sequentially: from the configuration of ship detection based on YOLOv8 to the development of transfer learning training configuration; from the DeepOCSORT configuration, identity-preserving tracking based on OSNet, to the inter-frame motion parameter estimation and the ingestion of the AIS stream; from Haversine-based sensor fusion to Kalman filter state smoothing. The network architecture, training process and sequence construction of features at runtime of a LSTM trajectory prediction module were described, as well as the kinematic fallback strategy for short-history tracks. Last, the pipeline's end-to-end components, the land mask filtering system and the dual-output visualisation system (OpenCV-annotated video feeds and Folium-created interactive geospatial maps) were displayed. The implementation described in this chapter is used to implement the entire surveillance framework whose performance is assessed and analysed in the following chapter.



Chapter 5

Results & Discussions

CHAPTER 5

RESULTS & DISCUSSIONS

Every chapter should start with an introduction paragraph. This paragraph should brief about the flow of the chapter. This introduction can be limited within 4 to 5 sentences. The chapter heading should be appropriately modified (a sample heading is shown for this chapter). But don't start the introduction paragraph in the chapters like "This chapter deals with...". Instead you should bring in the highlights of the chapter in the introduction paragraph.

5.1 Contents of this chapter

All the results obtained for your objectives should be discussed in this chapter. This chapter should contain the following sections as per the project.

1. Simulation results
2. Experimental results
3. Performance Comparison
4. Inferences drawn from the results obtained

All the figures should be properly explained by bringing the scenarios of the design done in the project. A detailed discussion of results obtained should be done in this chapter.

5.2 Tables in thesis

- All Table Caption should be in Sentence Case, TNR 10 Pt. It should be of the Format:
 - Table 1.1 Results of the experiment ... (Centered)
- It should be cited as Table 1.1.
- Caption should appear above the Table.
- Table Header and the entries should be of Font TNR 10 Pt, Justified.
- For wider Table, the page orientation can be Landscape.

- For Larger Table, it can run to pages and the header should be repeated for each page of the Table.
- Table must be adjusted to fit in the page and no single row is left out for a new page.

Sample Table 5.1 is given below for your reference,

Table 5.1: Country List

Country Name or Area Name	ISO ALPHA 2 Code	ISO ALPHA 3 Code	ISO numeric Code
Afghanistan	AF	AFG	004
Aland Islands	AX	ALA	248
Albania	AL	ALB	008
Algeria	DZ	DZA	012
American Samoa	AS	ASM	016
Andorra	AD	AND	020
Angola	AO	AGO	024

5.3 Math equation in thesis

All equation should be written using equation editor or using an equivalent tool.

- Equations should be numbered as : 1.1, 1.2 ...
- Equation should be Centered, 12 Pt, TNR.
- Equation number should be right Justified
- It should be cited as Eqn. 1.1.
- If the sentence starts by citing an equation, then it should be written as Equation 1.1 For example, Equation 5.1 states the Pythagoras theorem.

For example in Eqn. 5.1, The well known Pythagorean theorem $x^2 + y^2 = z^2$ was proved to be invalid for other exponents. Meaning the next equation has no integer solutions:

$$x^n + y^n = z^n \quad (5.1)$$

The mass-energy equivalence is described by the famous equation in Eqn. 5.2

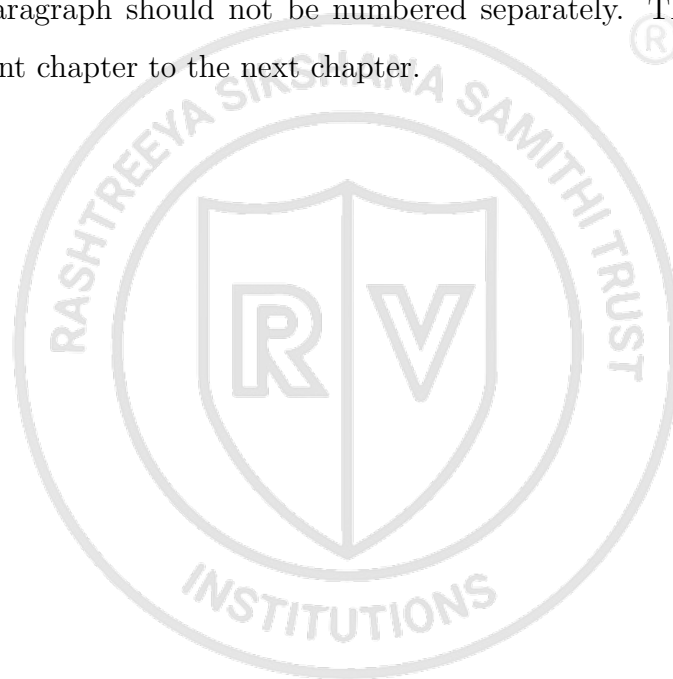
$$E = mc^2 \quad (5.2)$$

discovered in 1905 by Albert Einstein.

5.4 Interpretation of results

The chapters should not end with figures, instead bring the paragraph explaining about the figure at the end followed by a summary paragraph.

After elaborating the various sections of the chapter (From Chapter 2 onwards), a summary paragraph should be written discussing the highlights of that particular chapter. This summary paragraph should not be numbered separately. This paragraph should connect the present chapter to the next chapter.





Chapter 6

Conclusion and Future Scope

CHAPTER 6

CONCLUSION AND FUTURE SCOPE

Every chapter should start with an introduction paragraph. This paragraph should brief about the flow of the chapter. This introduction can be limited within 4 to 5 sentences. The chapter heading should be appropriately modified (a sample heading is shown for this chapter). But don't start the introduction paragraph in the chapters like "This chapter deals with...". Instead you should bring in the highlights of the chapter in the introduction paragraph.

6.1 Conclusion

This chapter should not contain an introduction paragraph like other chapters. You can directly write conclusion of the work done under this section. Typically this section can have 3 to 4 paragraphs.

First paragraph should bring in the scenario of the project and every objective should be explained here.

Second paragraph should say how the objectives are implemented and achieved.

Last paragraph should draw the conclusions from each objective with quantitative results, performance improvement etc.

6.2 Future Scope

Briefly discuss the constraints and limitations of the project and state the possibilities of extending the work in future.

6.3 Learning Outcomes of the Project

- List the learning outcomes here
- List a minimum of 5 learning outcomes



Appendix A

Code

APPENDIX A

CODE

A.1 First Appendix

You can use `tcbllisting` for creating the code snippets. The following example illustrates how one can customize the `tcbllisting` to achieve the `tcl` script. Similarly, one can use it for other programming language listing, including HDL.

```
# Since our design has a clock with name clk,
## specify that name under [get_port ]
create_clock -period 40 -waveform {0 20} [get_ports clk]

# Setting a 'delay' on the clock:
set_clock_latency 0.3 clk

# Setting up constraints on your I/P and O/P pins
set_input_delay 2.0 -clock clk [all_inputs]
set_output_delay 1.65 -clock clk [all_outputs]

# Set realistic 'loads' on each output pin
set_load 0.1 [all_outputs]

# Set 'maximum' fanin and fan-out for the input and output pins
set_max_fanout 1 [all_inputs]
set_fanout_load 8 [all_outputs]
```